APM5557: Partial Differential Equations
Homework 2 Problems

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Problem 1. From problem 4.1-3 in the book

Prove that the solution of the Robin or third boundary value problem for the Laplace equation:
$\Delta u = 0$ in Ω, with $\frac{\partial u}{\partial n} + \alpha u = \beta$ on $\partial\Omega$
is unique when $\alpha \geq 0$ is a constant.

Solution. Suppose u_1 and u_2 are two solutions to the Robin boundary value problem. Then their difference $v = u_1 - u_2$ satisfies the Laplace equation and the homogeneous Robin boundary condition:

$\Delta v = 0$ in Ω and $\frac{\partial v}{\partial n} + \alpha v = 0$ on $\partial\Omega$.

Now we use Green's first identity for v and v, which states that for a sufficiently smooth function v on a domain Ω with boundary $\partial\Omega$:

$$\int_\Omega \left(v\Delta v + |\nabla v|^2\right) dx = \int_{\partial\Omega} v \frac{\partial v}{\partial n}\, dS. \tag{1}$$

Now, since $\Delta v = 0$, the left-hand side simplifies to:

$$\int_\Omega |\nabla v|^2\, dx = \int_{\partial\Omega} v \frac{\partial v}{\partial n}\, dS. \tag{2}$$

From the Robin boundary condition, we have $\frac{\partial v}{\partial n} = -\alpha v$ on $\partial\Omega$. Substituting this into the boundary integral gives:

$$\int_\Omega |\nabla v|^2\, dx = \int_{\partial\Omega} v(-\alpha v)\, dS = -\alpha \int_{\partial\Omega} v^2\, dS. \tag{3}$$

If $\alpha \geq 0$, then the right-hand side is non-positive, while the left-hand side is non-negative.

This means that $\nabla v = 0$ in Ω and $\alpha v = 0$ and $v = 0$ on $\partial\Omega$

Since $\nabla v = 0$ in Ω, v is constant in Ω, and since $v = 0$ on the boundary, it must be that $v = 0$ everywhere in Ω. □

Problem 2. From problem 4.1-7 in the book

Solution. Solution goes here. □

Problem 3. From problem 4.1-9a in the book

Solution. Solution goes here. □